🤖 NewsBot Intelligence System

ITAI 2373 - Mid-Term Group Project Template

Team Members: [Add your names here] **Date:** [Add date] **GitHub Repository:** [Add your repo URL here]

🎯 Project Overview

Welcome to your NewsBot Intelligence System! This notebook will guide you through building a comprehensive NLP system that:

- 📰 **Processes** news articles with advanced text cleaning
- 🏷️ **Classifies** articles into categories (Politics, Sports, Technology, Business, Entertainment, Health)
- 🔍 **Extracts** named entities (people, organizations, locations, dates, money)
- 😊 **Analyzes** sentiment and emotional tone
- 📊 **Generates** insights for business intelligence

📚 Module Integration Checklist

- ☐ **Module 1:** NLP applications and real-world context
- ☐ **Module 2:** Text preprocessing pipeline
- ☐ **Module 3:** TF-IDF feature extraction
- ☐ **Module 4:** POS tagging analysis
- ☐ **Module 5:** Syntax parsing and semantic analysis
- ☐ **Module 6:** Sentiment and emotion analysis
- ☐ **Module 7:** Text classification system
- ☐ **Module 8:** Named Entity Recognition

```
from google.colab import files
uploaded = files.upload()
import pandas as pd

# Load the dataset
df = pd.read_csv("BBC News Train.csv")

# Basic information
print("Dataset shape:", df.shape)
```

```
print("\nColumns:")
print(df.columns)

print("\nCategory distribution:")
print(df["Category"].value_counts())

# Display first rows
df.head()
```

Choose Files BBC News Train.csv

BBC News Train.csv(text/csv) - 3351206 bytes, last modified: 6/23/2026 - 100% done

```
Saving BBC News Train.csv to BBC News Train (1).csv
Dataset shape: (1490, 3)

Columns:
Index(['ArticleId', 'Text', 'Category'], dtype='object')

Category distribution:
Category
sport            346
business         336
politics         274
entertainment    273
tech             261
Name: count, dtype: int64
```

	ArticleId	Text	Category
0	1833	worldcom ex-boss launches defence lawyers defe...	business
1	154	german business confidence slides german busin...	business
2	1101	bbc poll indicates economic gloom citizens in ...	business
3	1976	lifestyle governs mobile choice faster bett...	tech
4	917	enron bosses in $168m payout eighteen former e...	business

Next steps: Generate code with df | New interactive sheet

```
print(df.isnull().sum())
```

```
ArticleId    0
Text         0
Category     0
dtype: int64
```

📦 Setup and Installation

Let's start by installing and importing all the libraries we'll need for our NewsBot system.

```
# Install required packages (run this cell first!)
!pip install spacy scikit-learn nltk pandas matplotlib seaborn wordcloud plotly
!python -m spacy download en_core_web_sm

# Download NLTK data
import nltk
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('vader_lexicon')
nltk.download('averaged_perceptron_tagger')

print("✅ All packages installed successfully!")
```

```
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/di
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dis
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/d
```

```
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package vader_lexicon to /root/nltk_data...
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
✅ All packages installed successfully!
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger.zip.
```

```
# Import all necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud
import plotly.express as px
import plotly.graph_objects as go
from collections import Counter, defaultdict
import re
import warnings
warnings.filterwarnings('ignore')

# NLP Libraries
import spacy
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.stem import WordNetLemmatizer
from nltk.sentiment import SentimentIntensityAnalyzer
from nltk.tag import pos_tag

# Scikit-learn for machine learning
from sklearn.feature_extraction.text import TfidfVectorizer, CountVectorizer
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.naive_bayes import MultinomialNB
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, accuracy_s
from sklearn.pipeline import Pipeline

# Load spaCy model
nlp = spacy.load('en_core_web_sm')

# Set up plotting style
```

```
plt.style.use('default')
sns.set_palette("husl")

print("📚 All libraries imported successfully!")
print(f"🔧 spaCy model loaded: {nlp.meta['name']} v{nlp.meta['version']}")
```

```
📚 All libraries imported successfully!
🔧 spaCy model loaded: core_web_sm v3.8.0
```

📊 Data Loading and Exploration

🎯 Module 1: Understanding Our NLP Application

Before we dive into the technical implementation, let's understand the real-world context of our NewsBot Intelligence System. This system addresses several business needs:

1. **Media Monitoring:** Automatically categorize and track news coverage
2. **Business Intelligence:** Extract key entities and sentiment trends
3. **Content Management:** Organize large volumes of news content
4. **Market Research:** Understand public sentiment about topics and entities

💡 **Discussion Question:** What other real-world applications can you think of for this type of system? Consider different industries and use cases.

```
# Load the BBC News dataset
import pandas as pd

# Read the CSV file uploaded to Colab
df = pd.read_csv("BBC News Train.csv")

# Rename columns to match the notebook's expected format
df = df.rename(columns={
    "ArticleId": "article_id",
    "Text": "content",
    "Category": "category"
})

# Create a title column from the first few words of the article
df["title"] = df["content"].str.split().str[:8].str.join("...")

# Add placeholder columns expected by the notebook
df["date"] = "2024-01-15"
df["source"] = "BBC News"

# Rearrange columns
df = df[["article_id", "title", "content", "category", "date", "source"]]
```

```
print("📊 Dataset loaded successfully!")
print(f"📈 Shape: {df.shape}")
print(f"📋 Columns: {list(df.columns)}")

# Display first few rows
df.head()
```

```
📊 Dataset loaded successfully!
📈 Shape: (1490, 6)
📋 Columns: ['article_id', 'title', 'content', 'category', 'date', 'source']
```

	article_id	title	content	category	dat
0	1833	worldcom...ex-boss...launches...defence...lawy...	worldcom ex-boss launches defence lawyers defe...	business	202 01-1
1	154	german...business...confidence...slides...germ...	german business confidence slides german busin...	business	202 01-1
2	1101	bbc...poll...indicates...economic...gloom...ci...	bbc poll indicates economic gloom citizens in ...	business	202 01-1
3	1976	lifestyle...governs...mobile...choice...faster...	lifestyle governs mobile choice faster bett...	tech	202 01-1
4	917	enron...bosses...in...$168m...payout...eightee...	enron bosses in $168m payout eighteen former e...	business	202 01-1

Next steps: Generate code with df | New interactive sheet

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Check for missing values
print("\n🔍 MISSING VALUES")
print("=" * 50)
print(df.isnull().sum())

# Analyze article length
df['text_length'] = df['content'].str.len()
df['word_count'] = df['content'].str.split().str.len()

print("\n📝 ARTICLE LENGTH STATISTICS")
print("=" * 50)
print(df['word_count'].describe())

# Visualize word counts
plt.figure(figsize=(10,6))
sns.histplot(df['word_count'], bins=30)
plt.title('Distribution of Article Word Counts')
plt.xlabel('Number of Words')
plt.ylabel('Number of Articles')
plt.show()

# Source distribution
print("\n📰 SOURCE DISTRIBUTION")
print("=" * 50)
print(df['source'].value_counts())

# Data quality checks
print("\n✅ DATA QUALITY CHECK")
print("=" * 50)
print(f"Duplicate articles: {df.duplicated(subset=['content']).sum()}")
print(f"Empty articles: {(df['content'].str.len() == 0).sum()}")
print(f"Shortest article: {df['word_count'].min()} words")
print(f"Longest article: {df['word_count'].max()} words")
```

```
🔍 MISSING VALUES
==================================================
article_id      0
title           0
content         0
category        0
date            0
source          0
dtype: int64

📝 ARTICLE LENGTH STATISTICS
==================================================
count    1490.000000
mean      389.012752
std       210.898616
min        90.000000
25%       239.000000
50%       337.000000
75%       468.750000
max      3345.000000
Name: word_count, dtype: float64
```

🧹 Text Preprocessing Pipeline

🎯 Module 2: Advanced Text Preprocessing

Now we'll implement a comprehensive text preprocessing pipeline that cleans and normalizes our news articles. This is crucial for all downstream NLP tasks.

Key Preprocessing Steps:

1. **Text Cleaning:** Remove HTML, URLs, special characters
2. **Tokenization:** Split text into individual words
3. **Normalization:** Convert to lowercase, handle contractions
4. **Stop Word Removal:** Remove common words that don't carry meaning
5. **Lemmatization:** Reduce words to their base form

💡 **Think About:** Why is preprocessing so important? What happens if we skip these steps?

```
import nltk
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')

from nltk.stem import WordNetLemmatizer
from nltk.corpus import stopwords
import re
import pandas as pd
from collections import Counter

# Initialize preprocessing tools
lemmatizer = WordNetLemmatizer()
stop_words = set(stopwords.words('english'))

def clean_text(text):
    """
    Comprehensive text cleaning function.
    Removes HTML, URLs, emails, special characters, numbers, and extra spaces.
    """
    if pd.isna(text):
        return ""

    text = str(text).lower()

    # Remove HTML tags
    text = re.sub(r'<[^>]+>', '', text)
```

```
    # Remove URLs
    text = re.sub(r'http\S+|www\S+|https\S+', '', text)

    # Remove email addresses
    text = re.sub(r'\S+@\S+', '', text)

    # Remove special characters and digits
    text = re.sub(r'[^a-zA-Z\s]', '', text)

    # Remove extra whitespace
    text = re.sub(r'\s+', ' ', text).strip()

    return text

def preprocess_text(text, remove_stopwords=True, lemmatize=True):
    """
    Complete preprocessing pipeline:
    cleaning, tokenization, stop word removal, lemmatization.
    """
    text = clean_text(text)

    if not text:
        return ""

    # Simple tokenization after cleaning
    tokens = text.split()

    # Remove stop words
    if remove_stopwords:
        tokens = [token for token in tokens if token not in stop_words]

    # Lemmatize
    if lemmatize:
        tokens = [lemmatizer.lemmatize(token) for token in tokens]

    # Remove very short words
    tokens = [token for token in tokens if len(token) > 2]

    return ' '.join(tokens)

# Test the preprocessing function
sample_text = "Apple Inc. announced record quarterly earnings today! Visit http

print("Original text:")
print(sample_text)

print("\nCleaned text:")
print(clean_text(sample_text))
```

```
print("\nFully preprocessed text:")
print(preprocess_text(sample_text))
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
Original text:
Apple Inc. announced record quarterly earnings today! Visit https://apple.com fo

Cleaned text:
apple inc announced record quarterly earnings today visit for more info technews

Fully preprocessed text:
apple inc announced record quarterly earnings today visit info technews
```

```
# Apply preprocessing to the dataset
print("🧹 Preprocessing all articles...")

# Make sure required columns exist
if 'title' not in df.columns:
    df['title'] = df['content'].str.split().str[:8].str.join(' ')

if 'content' not in df.columns:
    raise ValueError("The dataframe must contain a 'content' column.")

# Fill missing text values just in case
df['title'] = df['title'].fillna("")
df['content'] = df['content'].fillna("")

# Create new columns for processed text
df['title_clean'] = df['title'].apply(clean_text)
df['content_clean'] = df['content'].apply(clean_text)
df['title_processed'] = df['title'].apply(preprocess_text)
df['content_processed'] = df['content'].apply(preprocess_text)

# Combine title and content for full article analysis
df['full_text'] = df['title'].astype(str) + ' ' + df['content'].astype(str)
df['full_text_processed'] = df['full_text'].apply(preprocess_text)

print("✅ Preprocessing complete!")

# Show before and after examples
print("\n📝 BEFORE AND AFTER EXAMPLES")
print("=" * 60)
for i in range(min(3, len(df))):
    print(f"\nExample {i+1}:")
    print(f"Original: {df.iloc[i]['full_text'][:150]}...")
    print(f"Processed: {df.iloc[i]['full_text_processed'][:150]}...")
```

```
# Analyze preprocessing results
print("\n📊 PREPROCESSING ANALYSIS")
print("=" * 60)

df['original_word_count'] = df['full_text'].str.split().str.len()
df['processed_word_count'] = df['full_text_processed'].str.split().str.len()

print(f"Average original word count: {df['original_word_count'].mean():.2f}")
print(f"Average processed word count: {df['processed_word_count'].mean():.2f}")

original_words = ' '.join(df['full_text'].astype(str).str.lower()).split()
processed_words = ' '.join(df['full_text_processed'].astype(str)).split()

print(f"Unique words before preprocessing: {len(set(original_words))}")
print(f"Unique words after preprocessing: {len(set(processed_words))}")

common_words = Counter(processed_words).most_common(20)

print("\nMost common words after preprocessing:")
for word, count in common_words:
    print(f"{word}: {count}")
```

```
🧹 Preprocessing all articles...
✅ Preprocessing complete!

📝 BEFORE AND AFTER EXAMPLES
============================================================

Example 1:
Original: worldcom...ex-boss...launches...defence...lawyers...defending...former
Processed: worldcomexbosslaunchesdefencelawyersdefendingformerworldcom worldcom

Example 2:
Original: german...business...confidence...slides...german...business...confiden
Processed: germanbusinessconfidenceslidesgermanbusinessconfidencefell german bus

Example 3:
Original: bbc...poll...indicates...economic...gloom...citizens...in...a bbc poll
Processed: bbcpollindicateseconomicgloomcitizensina bbc poll indicates economic

📊 PREPROCESSING ANALYSIS
============================================================
Average original word count: 386.01
Average processed word count: 208.08
Unique words before preprocessing: 37025
Unique words after preprocessing: 23919

Most common words after preprocessing:
said: 4838
year: 1872
would: 1711
also: 1426
new: 1334
people: 1323
```

```
one: 1190
could: 1032
game: 949
time: 940
first: 893
last: 883
say: 844
two: 816
world: 811
film: 802
government: 771
make: 695
company: 682
firm: 675
```

📊 Feature Extraction and Statistical Analysis

🎯 Module 3: TF-IDF Analysis

Now we'll extract numerical features from our text using TF-IDF (Term Frequency-Inverse Document Frequency). This technique helps us identify the most important words in each document and across the entire corpus.

TF-IDF Key Concepts:

- **Term Frequency (TF):** How often a word appears in a document
- **Inverse Document Frequency (IDF):** How rare a word is across all documents
- **TF-IDF Score:** TF × IDF - balances frequency with uniqueness

💡 **Business Value:** TF-IDF helps us identify the most distinctive and important terms for each news category.

```
# Create TF-IDF vectorizer
# 💡 TIP: Experiment with different parameters:
# - max_features: limit vocabulary size
# - ngram_range: include phrases (1,1) for words, (1,2) for words+bigrams
# - min_df: ignore terms that appear in less than min_df documents
# - max_df: ignore terms that appear in more than max_df fraction of documents
from sklearn.feature_extraction.text import TfidfVectorizer

tfidf_vectorizer = TfidfVectorizer(
    max_features=5000,  # Limit vocabulary for computational efficiency
    ngram_range=(1, 2),  # Include unigrams and bigrams
    min_df=2,  # Ignore terms that appear in less than 2 documents
    max_df=0.8  # Ignore terms that appear in more than 80% of documents
)

# Fit and transform the processed text
```

```
print("🔢 Creating TF-IDF features...")
tfidf_matrix = tfidf_vectorizer.fit_transform(df['full_text_processed'])
feature_names = tfidf_vectorizer.get_feature_names_out()

print(f"✅ TF-IDF matrix created!")
print(f"📊 Shape: {tfidf_matrix.shape}")
print(f"📝 Vocabulary size: {len(feature_names)}")
print(f"🔢 Sparsity: {(1 - tfidf_matrix.nnz / (tfidf_matrix.shape[0] * tfidf_m

# Convert to DataFrame for easier analysis
tfidf_df = pd.DataFrame(tfidf_matrix.toarray(), columns=feature_names)
tfidf_df['category'] = df['category'].values

print("\n🔍 Sample TF-IDF features:")
print(tfidf_df.iloc[:3, :10])  # Show first 3 rows and 10 features
```

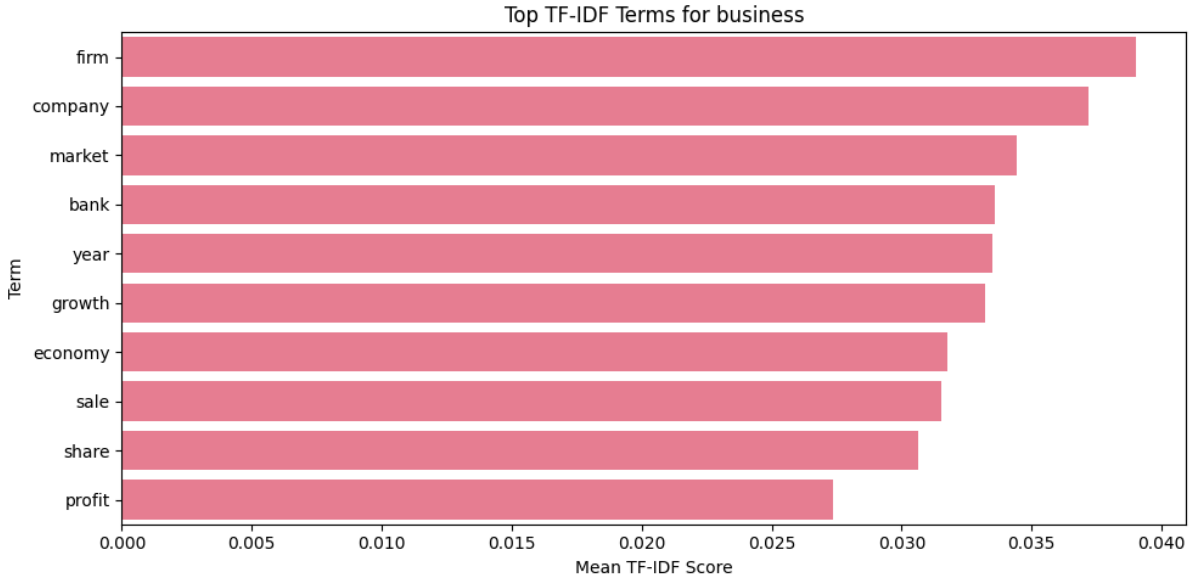
```
🔢 Creating TF-IDF features...
✅ TF-IDF matrix created!
📊 Shape: (1490, 5000)
📝 Vocabulary size: 5000
🔢 Sparsity: 97.50%

🔍 Sample TF-IDF features:
   abc  ability  able  abroad  absence  absolute  absolutely  abuse  abused  \
0  0.0      0.0   0.0     0.0      0.0       0.0         0.0    0.0     0.0
1  0.0      0.0   0.0     0.0      0.0       0.0         0.0    0.0     0.0
2  0.0      0.0   0.0     0.0      0.0       0.0         0.0    0.0     0.0

   academy
0      0.0
1      0.0
2      0.0
```

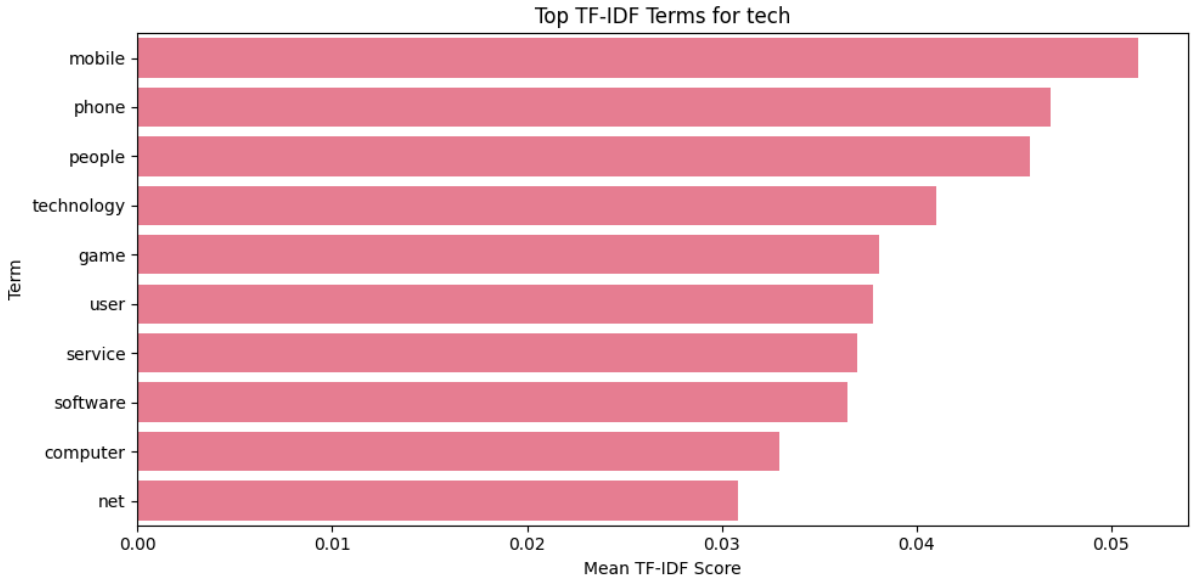
```
# Analyze most important terms per category
def get_top_tfidf_terms(category, n_terms=10):
    """
    Get top TF-IDF terms for a specific category.
    """
    category_data = tfidf_df[tfidf_df['category'] == category]
    mean_scores = category_data.drop('category', axis=1).mean().sort_values(asc
    return mean_scores.head(n_terms)

# Analyze top terms for each category
print("🏷️ TOP TF-IDF TERMS BY CATEGORY")
print("=" * 50)

categories = df['category'].unique()
category_terms = {}

for category in categories:
    top_terms = get_top_tfidf_terms(category, n_terms=10)
    category_terms[category] = top_terms
```

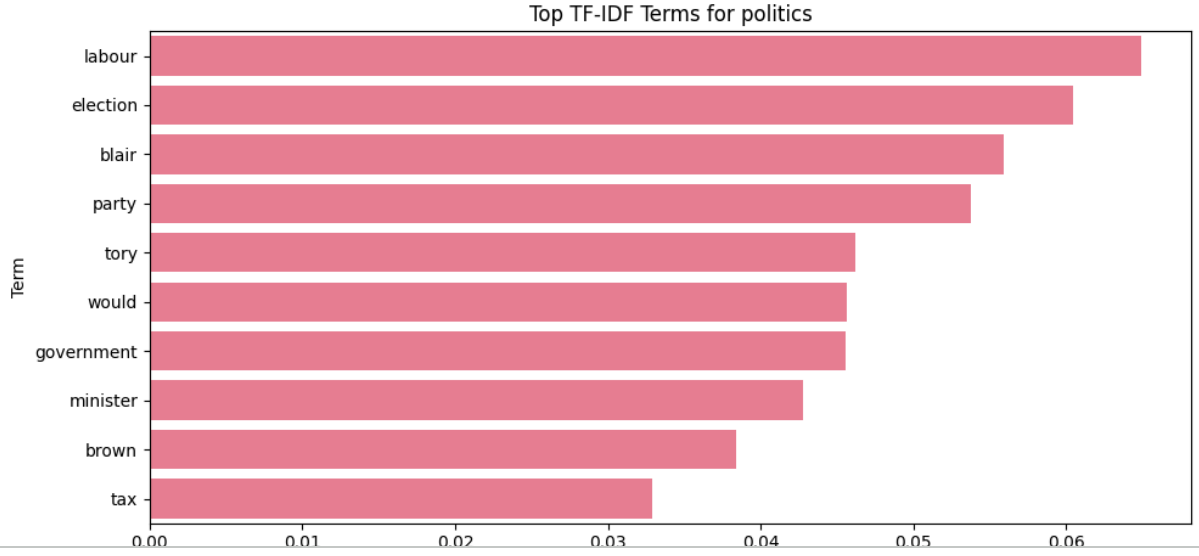
```
    print(f"\n {category.upper()}:")
    for term, score in top_terms.items():
        print(f"  {term}: {score:.4f}")

# Bar charts of top terms by category
for category, terms in category_terms.items():
    plt.figure(figsize=(10, 5))
    sns.barplot(x=terms.values, y=terms.index)
    plt.title(f"Top TF-IDF Terms for {category}")
    plt.xlabel("Mean TF-IDF Score")
    plt.ylabel("Term")
    plt.tight_layout()
    plt.show()

# Heatmap of top term importance across categories
top_all_terms = list(set([term for terms in category_terms.values() for term ir

heatmap_data = []

for category in categories:
    category_data = tfidf_df[tfidf_df['category'] == category]
    mean_scores = category_data.drop('category', axis=1).mean()
    heatmap_data.append([mean_scores.get(term, 0) for term in top_all_terms])

heatmap_df = pd.DataFrame(heatmap_data, index=categories, columns=top_all_terms

plt.figure(figsize=(14, 6))
sns.heatmap(heatmap_df, cmap="YlGnBu", annot=False)
plt.title("TF-IDF Term Importance Across Categories")
plt.xlabel("Terms")
plt.ylabel("Category")
plt.tight_layout()
plt.show()
```

```
🏷️ TOP TF-IDF TERMS BY CATEGORY
==================================================

📰 BUSINESS:
  firm: 0.0390
  company: 0.0372
  market: 0.0344
  bank: 0.0336
  year: 0.0335
  growth: 0.0332
  economy: 0.0317
  sale: 0.0316
  share: 0.0307
  profit: 0.0273

📰 TECH:
  mobile: 0.0514
  phone: 0.0469
  people: 0.0458
  technology: 0.0410
  game: 0.0381
  user: 0.0378
  service: 0.0370
  software: 0.0365
  computer: 0.0330
  net: 0.0308

📰 POLITICS:
  labour: 0.0649
  election: 0.0605
  blair: 0.0559
  party: 0.0538
  tory: 0.0462
  would: 0.0456
  government: 0.0455
  minister: 0.0428
  brown: 0.0384
  tax: 0.0329

📰 SPORT:
  game: 0.0448
  england: 0.0374
  win: 0.0341
  player: 0.0322
  match: 0.0305
  champion: 0.0293
  cup: 0.0278
  team: 0.0259
  chelsea: 0.0255
  injury: 0.0249

📰 ENTERTAINMENT:
  film: 0.0995
  award: 0.0524
  best: 0.0460
  show: 0.0376
```

```
star: 0.0372
music: 0.0356
band: 0.0349
actor: 0.0335
year: 0.0300
album: 0.0290
```







🏷️ Part-of-Speech Analysis

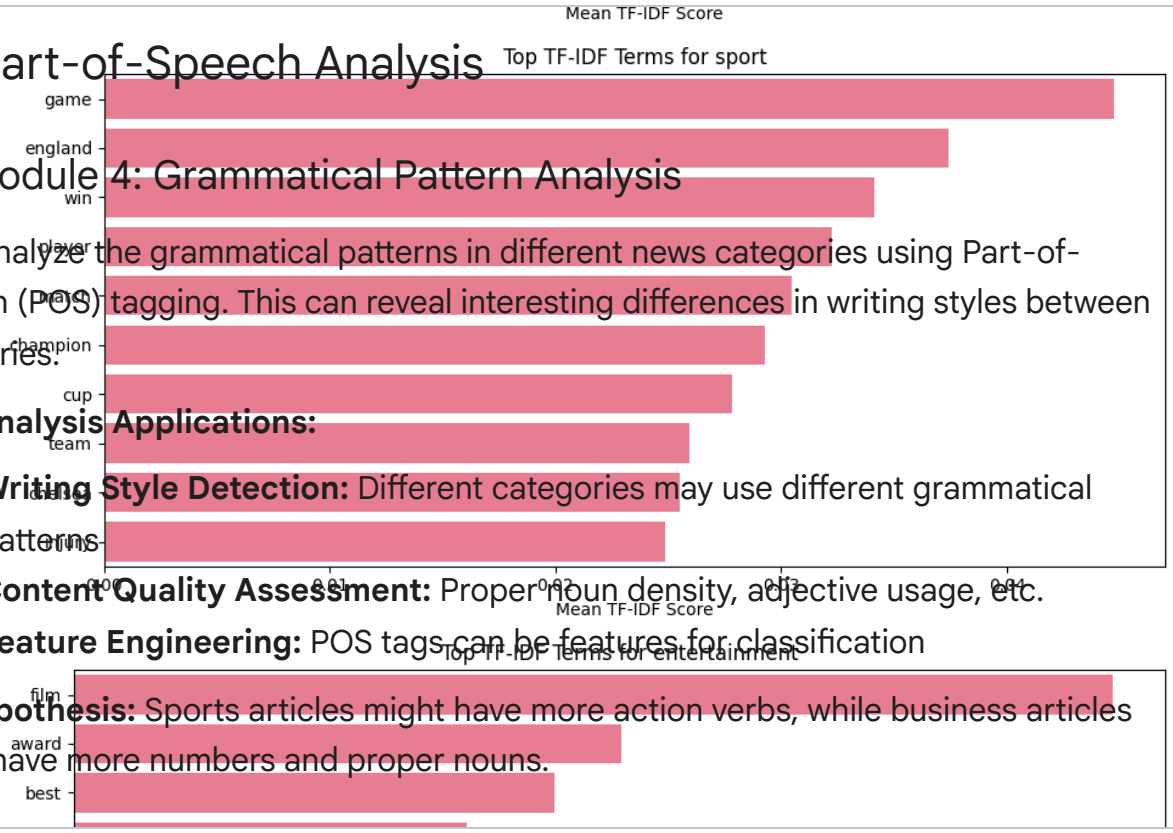
🎯 Module 4: Grammatical Pattern Analysis

Let's analyze the grammatical patterns in different news categories using Part-of-Speech (POS) tagging. This can reveal interesting differences in writing styles between categories.

POS Analysis Applications:

- **Writing Style Detection:** Different categories may use different grammatical patterns
- **Content Quality Assessment:** Proper noun density, adjective usage, etc.
- **Feature Engineering:** POS tags can be features for classification

💡 **Hypothesis:** Sports articles might have more action verbs, while business articles might have more numbers and proper nouns.




```
import nltk
nltk.download('averaged_perceptron_tagger_eng')

from nltk import pos_tag
from collections import Counter

def analyze_pos_patterns(text):
    """
    Analyze POS patterns in text.
    Uses simple split tokenization to avoid NLTK punkt issues.
    """
    if not text or pd.isna(text):
        return {}

    # Clean and tokenize text
    cleaned = clean_text(text)
    tokens = cleaned.split()

    if len(tokens) == 0:
        return {}

    # POS tagging
    pos_tags = pos_tag(tokens)

    # Count POS categories
    pos_counts = Counter([tag for word, tag in pos_tags])
    total_words = len(pos_tags)

    # Convert counts to proportions
```

```
    pos_proportions = {pos: count / total_words for pos, count in pos_counts.it

    return pos_proportions

# Apply POS analysis to all articles
print("🔍 Analyzing POS patterns...")

pos_results = []

for idx, row in df.iterrows():
    pos_analysis = analyze_pos_patterns(row['full_text'])
    pos_analysis['category'] = row['category']
    pos_analysis['article_id'] = row['article_id']
    pos_results.append(pos_analysis)

# Convert to DataFrame
pos_df = pd.DataFrame(pos_results).fillna(0)

print("✅ POS analysis complete!")
print(f"📊 Found {len(pos_df.columns) - 2} different POS tags")

# Show sample results
print("\n📝 Sample POS analysis:")
display(pos_df.head())
```

```
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
🔍 Analyzing POS patterns...
✅ POS analysis complete!
📊 Found 37 different POS tags

📝 Sample POS analysis:
```

	NN	VBZ	NNS	VBG	JJ	IN	DT	VBP	
0	0.257627	0.023729	0.122034	0.030508	0.091525	0.118644	0.071186	0.023729	0
1	0.243671	0.015823	0.075949	0.034810	0.126582	0.145570	0.088608	0.015823	0
2	0.202454	0.030675	0.089980	0.030675	0.083845	0.137014	0.106339	0.020450	0
3	0.150641	0.022436	0.115385	0.025641	0.089744	0.131410	0.067308	0.049679	0
4	0.248571	0.025714	0.088571	0.040000	0.097143	0.134286	0.111429	0.017143	0

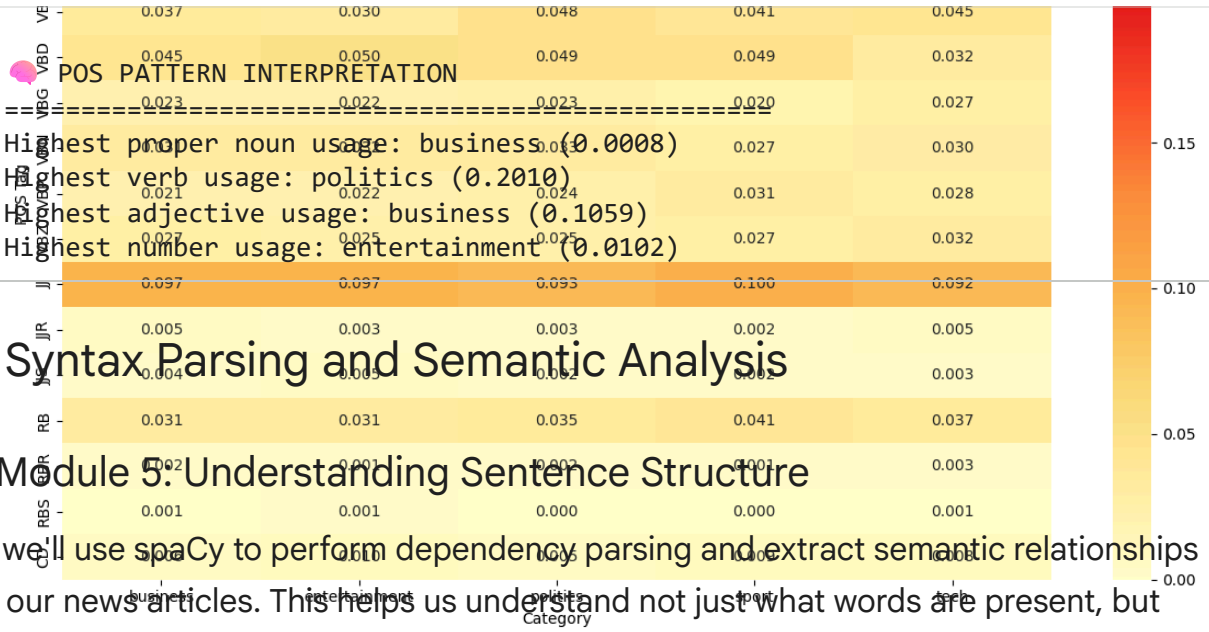
5 rows × 39 columns

```
# Analyze POS patterns by category
print("📊 POS PATTERNS BY CATEGORY")
print("=" * 50)

# Group by category and calculate mean proportions
```

```
pos_by_category = pos_df.groupby('category').mean()

# Focus on major POS categories
major_pos = ['NN', 'NNS', 'NNP', 'NNPS', 'VB', 'VBD', 'VBG', 'VBN', 'VBP', 'VBZ
             'JJ', 'JJR', 'JJS', 'RB', 'RBR', 'RBS', 'CD']

# Filter to only include major POS tags that exist in our data
available_pos = [pos for pos in major_pos if pos in pos_by_category.columns]

if available_pos:
    pos_summary = pos_by_category[available_pos]

    print("\n🎯 Key POS patterns by category:")
    print(pos_summary.round(4))

    # Create visualization
    plt.figure(figsize=(12, 8))
    sns.heatmap(pos_summary.T, annot=True, cmap='YlOrRd', fmt='.3f')
    plt.title('POS Tag Proportions by News Category')
    plt.xlabel('Category')
    plt.ylabel('POS Tag')
    plt.tight_layout()
    plt.show()

    # 💡 STUDENT TASK: Analyze the patterns
    # - Which categories use more nouns vs verbs?
    # - Do business articles have more numbers (CD)?
    # - Are there differences in adjective usage?

    print("\n💡 ANALYSIS QUESTIONS:")
    print("1. Which category has the highest proportion of proper nouns (NNP/NN
    print("2. Which category uses the most action verbs (VB, VBD, VBG)?")
    print("3. Are there interesting patterns in adjective (JJ) usage?")
    print("4. How does number (CD) usage vary across categories?")
else:
    print("⚠️ No major POS tags found in the analysis. Check your POS tagging
```

```
📊 POS PATTERNS BY CATEGORY
==================================================
```

```
# Additional POS interpretation for student task

print("\n🧠 POS PATTERN INTERPRETATION")
print("=" * 50)

# Proper nouns
proper_noun_cols = [col for col in ['NNP', 'NNPS'] if col in pos_by_category.cc
if proper_noun_cols:
    proper_nouns = pos_by_category[proper_noun_cols].sum(axis=1)
    print(f"Highest proper noun usage: {proper_nouns.idxmax()} ({proper_nouns.m

# Action verbs
verb_cols = [col for col in ['VB', 'VBD', 'VBG', 'VBN', 'VBP', 'VBZ'] if col ir
if verb_cols:
    verbs = pos_by_category[verb_cols].sum(axis=1)
    print(f"Highest verb usage: {verbs.idxmax()} ({verbs.max():.4f})")

# Adjectives
adj_cols = [col for col in ['JJ', 'JJR', 'JJS'] if col in pos_by_category.colum
if adj_cols:
    adjectives = pos_by_category[adj_cols].sum(axis=1)
    print(f"Highest adjective usage: {adjectives.idxmax()} ({adjectives.max():.

# Numbers
if 'CD' in pos_by_category.columns:
    numbers = pos_by_category['CD']
    print(f"Highest number usage: {numbers.idxmax()} ({numbers.max():.4f})")
```




```
🧠 POS PATTERN INTERPRETATION
==================================================
Highest proper noun usage: business (0.0008)
Highest verb usage: politics (0.2010)
Highest adjective usage: business (0.1059)
Highest number usage: entertainment (0.0102)
```

🌳 Syntax Parsing and Semantic Analysis

🎯 Module 5: Understanding Sentence Structure

Now we'll use spaCy to perform dependency parsing and extract semantic relationships from our news articles. This helps us understand not just what words are present, but how they relate to each other.

```
💡 ANALYSIS QUESTIONS:
1. Which category has the highest proportion of proper nouns (NNP/NNPS)?
2. Which category uses the most action verbs (VB, VBD, VBG)?
```

Dependency Parsing Applications:

3. Are there interesting patterns in adjective (JJ) usage?
4. How does number (CD) usage vary across categories?

- **Relationship Extraction:** Find connections between entities
- **Event Detection:** Identify who did what to whom
- **Information Extraction:** Extract structured facts from unstructured text

💡 **Business Value:** Understanding sentence structure helps extract more precise information about events, relationships, and actions mentioned in news articles.

```
def extract_syntactic_features(text):
    """
    Extract syntactic features using spaCy dependency parsing

    💡 TIP: This function should extract:
    - Dependency relations
    - Subject-verb-object patterns
    - Noun phrases
    - Verb phrases
    """
    if not text or pd.isna(text):
        return {}

    # Process text with spaCy
    doc = nlp(str(text))

    features = {
        'num_sentences': len(list(doc.sents)),
        'num_tokens': len(doc),
        'dependency_relations': [],
        'noun_phrases': [],
        'verb_phrases': [],
        'subjects': [],
        'objects': []
    }

    # 🚀 YOUR CODE HERE: Extract syntactic features

    # Extract dependency relations
    for token in doc:
        if not token.is_space and not token.is_punct:
            features['dependency_relations'].append(token.dep_)

    # Extract noun phrases
    for chunk in doc.noun_chunks:
        features['noun_phrases'].append(chunk.text.lower())

    # Extract subjects and objects
    for token in doc:
        if token.dep_ in ['nsubj', 'nsubjpass']:  # Subjects
            features['subjects'].append(token.text.lower())
        elif token.dep_ in ['dobj', 'iobj', 'pobj']:  # Objects
```

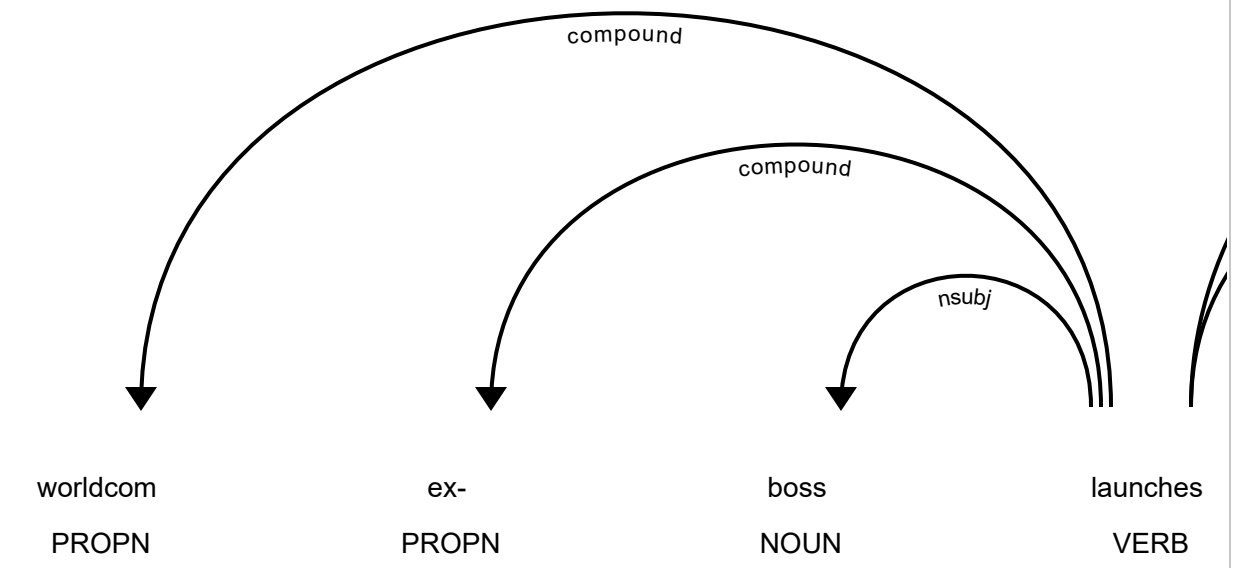
```
                features['objects'].append(token.text.lower())

        # Count dependency types
        dep_counts = Counter(features['dependency_relations'])
        features['dependency_counts'] = dict(dep_counts)

        return features

# Apply syntactic analysis to sample articles
print("🌳 Performing syntactic analysis...")

# Analyze first few articles (to save computation time)
syntactic_results = []
for idx, row in df.head(5).iterrows():  # Limit to first 5 for demo
    features = extract_syntactic_features(row['full_text'])
    features['category'] = row['category']
    features['article_id'] = row['article_id']
    syntactic_results.append(features)

print("✅ Syntactic analysis complete!")

# Display results
for i, result in enumerate(syntactic_results):
    print(f"\n📰 Article {i+1} ({result['category']}):")
    print(f"  Sentences: {result['num_sentences']}")
    print(f"  Tokens: {result['num_tokens']}")
    print(f"  Noun phrases: {result['noun_phrases'][:3]}...")  # Show first 3
    print(f"  Subjects: {result['subjects'][:3]}...")  # Show first 3
    print(f"  Objects: {result['objects'][:3]}...")  # Show first 3
```

```
🌳 Performing syntactic analysis...
✅ Syntactic analysis complete!

📰 Article 1 (business):
  Sentences: 15
  Tokens: 363
  Noun phrases: ['worldcom', 'ex-boss...launches', 'defence']...
  Subjects: ['worldcom', '-', 'boss']...
  Objects: ['ebbers', 'battery', 'charges']...

📰 Article 2 (business):
  Sentences: 15
  Tokens: 383
  Noun phrases: ['german', 'business', 'confidence']...
  Subjects: ['german', 'confidence', 'ifo']...
  Objects: ['confidence', 'february', 'hopes']...

📰 Article 3 (business):
  Sentences: 24
  Tokens: 602
  Noun phrases: ['bbc...poll', 'economic...gloom', 'citizens']...
  Subjects: ['poll', 'gloom', 'poll']...
  Objects: ['citizens', 'majority', 'nations']...
```

```
📰 Article 4 (tech):
  Sentences: 31
  Tokens: 739
  Noun phrases: ['lifestyle...governs', 'mobile...choice', '...funkier lifestyle
  Subjects: ['governs', 'firms', 'firms']...
  Objects: ['choice', 'hardware', 'research']...

📰 Article 5 (business):
  Sentences: 17
  Tokens: 432
  Noun phrases: ['enron', 'bosses', '$168m']...
  Subjects: ['enron', 'plaintiff', 'news']...
  Objects: ['$168m', 'payout', '168']...
```

```
# Visualize dependency parsing for a sample sentence
from spacy import displacy

# Choose a sample sentence
sample_sentence = df.iloc[0]['content']  # First article's content
print(f"📝 Sample sentence: {sample_sentence}")

# Process with spaCy
doc = nlp(sample_sentence)

# Display dependency tree (this works best in Jupyter)
print("\n🌳 Dependency Parse Visualization:")
try:
    # This will create an interactive visualization in Jupyter
    displacy.render(doc, style="dep", jupyter=True)
except:
    # Fallback: print dependency information
    print("\n🔗 Dependency Relations:")
    for token in doc:
        if not token.is_space and not token.is_punct:
            print(f"  {token.text} --> {token.dep_} --> {token.head.text}")

# 💡 STUDENT TASK: Extend syntactic analysis
# - Compare syntactic complexity across categories
# - Extract action patterns (who did what)
# - Identify most common dependency relations per category
# - Create features for classification based on syntax
```

```
📝 Sample sentence: worldcom ex-boss launches defence lawyers defending former
🌳 Dependency Parse Visualization:
```



😊 Sentiment and Emotion Analysis

🎯 Module 6: Understanding Emotional Tone

Let's analyze the sentiment and emotional tone of our news articles. This can reveal interesting patterns about how different types of news are presented and perceived.

Sentiment Analysis Applications:

- **Media Bias Detection:** Identify emotional slant in news coverage
- **Public Opinion Tracking:** Monitor sentiment trends over time
- **Content Recommendation:** Suggest articles based on emotional tone

💡 **Hypothesis:** Different news categories might have different emotional profiles - sports might be more positive, politics more negative, etc.

```
# Initialize sentiment analyzer
sia = SentimentIntensityAnalyzer()

def analyze_sentiment(text):
    """
    Analyze sentiment using VADER sentiment analyzer

    💡 TIP: VADER returns:
    - compound: overall sentiment (-1 to 1)
    - pos: positive score (0 to 1)
    - neu: neutral score (0 to 1)
    - neg: negative score (0 to 1)
    """
    if not text or pd.isna(text):
        return {'compound': 0, 'pos': 0, 'neu': 1, 'neg': 0}

    # 🚀 YOUR CODE HERE: Implement sentiment analysis
    scores = sia.polarity_scores(str(text))

    # Add interpretation
    if scores['compound'] >= 0.05:
        scores['sentiment_label'] = 'positive'
    elif scores['compound'] <= -0.05:
        scores['sentiment_label'] = 'negative'
    else:
        scores['sentiment_label'] = 'neutral'

    return scores

# Apply sentiment analysis to all articles
print("😊 Analyzing sentiment...")

sentiment_results = []
for idx, row in df.iterrows():
    # Analyze both title and content
    title_sentiment = analyze_sentiment(row['title'])
    content_sentiment = analyze_sentiment(row['content'])
    full_sentiment = analyze_sentiment(row['full_text'])
```

```
        result = {
            'article_id': row['article_id'],
            'category': row['category'],
            'title_sentiment': title_sentiment['compound'],
            'title_label': title_sentiment['sentiment_label'],
            'content_sentiment': content_sentiment['compound'],
            'content_label': content_sentiment['sentiment_label'],
            'full_sentiment': full_sentiment['compound'],
            'full_label': full_sentiment['sentiment_label'],
            'pos_score': full_sentiment['pos'],
            'neu_score': full_sentiment['neu'],
            'neg_score': full_sentiment['neg']
        }
        sentiment_results.append(result)

    # Convert to DataFrame
    sentiment_df = pd.DataFrame(sentiment_results)

    print("✅ Sentiment analysis complete!")
    print(f"📊 Analyzed {len(sentiment_df)} articles")

    # Display sample results
    print("\n📝 Sample sentiment results:")
    print(sentiment_df[['category', 'full_sentiment', 'full_label']].head())
```

```
😊 Analyzing sentiment...
✅ Sentiment analysis complete!
📊 Analyzed 1490 articles

📝 Sample sentiment results:
   category  full_sentiment full_label
0  business         -0.9701   negative
1  business          0.7623   positive
2  business         -0.9318   negative
3      tech          0.9554   positive
4  business         -0.9486   negative
```

```
    # Analyze sentiment patterns by category
    print("📊 SENTIMENT ANALYSIS BY CATEGORY")
    print("=" * 50)

    # Calculate sentiment statistics by category
    sentiment_by_category = sentiment_df.groupby('category').agg({
        'full_sentiment': ['mean', 'std', 'min', 'max'],
        'pos_score': 'mean',
        'neu_score': 'mean',
        'neg_score': 'mean'
    }).round(4)

    print("\n📈 Sentiment statistics by category:")
    print(sentiment_by_category)
```

```
# Sentiment distribution by category
sentiment_dist = sentiment_df.groupby(['category', 'full_label']).size().unstac
sentiment_dist_pct = sentiment_dist.div(sentiment_dist.sum(axis=1), axis=0) * 1

print("\n📊 Sentiment distribution (%) by category:")
print(sentiment_dist_pct.round(2))

# Create visualizations
fig, axes = plt.subplots(2, 2, figsize=(15, 12))

# 1. Sentiment scores by category
sns.boxplot(data=sentiment_df, x='category', y='full_sentiment', ax=axes[0,0])
axes[0,0].set_title('Sentiment Score Distribution by Category')
axes[0,0].tick_params(axis='x', rotation=45)

# 2. Sentiment label distribution
sentiment_dist_pct.plot(kind='bar', ax=axes[0,1], stacked=True)
axes[0,1].set_title('Sentiment Label Distribution by Category (%)')
axes[0,1].tick_params(axis='x', rotation=45)
axes[0,1].legend(title='Sentiment')

# 3. Positive vs Negative scores
category_means = sentiment_df.groupby('category')[['pos_score', 'neg_score']].m
category_means.plot(kind='bar', ax=axes[1,0])
axes[1,0].set_title('Average Positive vs Negative Scores by Category')
axes[1,0].tick_params(axis='x', rotation=45)
axes[1,0].legend(['Positive', 'Negative'])

# 4. Sentiment vs Category heatmap
sentiment_pivot = sentiment_df.pivot_table(values='full_sentiment', index='cate
                                           columns='full_label', aggfunc='count',
sns.heatmap(sentiment_pivot, annot=True, fmt='d', ax=axes[1,1], cmap='YlOrRd')
axes[1,1].set_title('Sentiment Count Heatmap')

plt.tight_layout()
plt.show()

# 💡 STUDENT TASK: Analyze sentiment patterns
# - Which categories are most positive/negative?
# - Are there differences between title and content sentiment?
# - How does sentiment vary within categories?
# - Can sentiment be used as a feature for classification?
```

```
📊 SENTIMENT ANALYSIS BY CATEGORY
==================================================

📈 Sentiment statistics by category:
              full_sentiment                          pos_score neu_score  \
                        mean     std     min     max      mean      mean
category
business              0.2404  0.8370 -0.9985  0.9995    0.0923    0.8375
entertainment         0.5991  0.6796 -0.9978  0.9999    0.1352    0.8112
politics              0.0578  0.8811 -0.9984  0.9999    0.0945    0.8188
sport                 0.5541  0.7244 -0.9967  0.9996    0.1356    0.7940
tech                  0.5213  0.7567 -0.9981  0.9993    0.0813    0.8691

              neg_score
                   mean
category
business         0.0702
entertainment    0.0536
politics         0.0867
sport            0.0704
tech             0.0506

📊 Sentiment distribution (%) by category:
full_label     negative  neutral  positive
category
business          36.31     1.49     62.20
entertainment     18.68     0.73     80.59
```

🏷️ Text Classification System

🎯 Module 7: Building the News Classifier

Now we'll build the core of our NewsBot system - a multi-class text classifier that can automatically categorize news articles. We'll compare different algorithms and evaluate their performance.

Classification Pipeline:

1. **Feature Engineering:** Combine TF-IDF with other features
2. **Model Training:** Train multiple algorithms
3. **Model Evaluation:** Compare performance metrics
4. **Model Selection:** Choose the best performing model

💡 **Business Impact:** Accurate classification enables automatic content routing, personalized recommendations, and efficient content management.

```
# Prepare features for classification
print("🔧 Preparing features for classification...")

# 💡 TIP: Combine multiple feature types for better performance
# - TF-IDF features (most important)
# - Sentiment features
# - Text length features
# - POS features (if available)

# Create feature matrix
X_tfidf = tfidf_matrix.toarray()  # TF-IDF features

# Add sentiment features
sentiment_features = sentiment_df[['full_sentiment', 'pos_score', 'neu_score', '

# Add text length features
length_features = np.array([
    df['full_text'].str.len(),  # Character length
    df['full_text'].str.split().str.len(),  # Word count
    df['title'].str.len(),  # Title length
]).T

# 🚀 YOUR CODE HERE: Combine all features
X_combined = np.hstack([
    X_tfidf,
    sentiment_features,
```

```
    length_features
])

# Target variable
y = df['category'].values

print(f"✅ Feature matrix prepared!")
print(f"📊 Feature matrix shape: {X_combined.shape}")
print(f"🎯 Number of classes: {len(np.unique(y))}")
print(f"📋 Classes: {np.unique(y)}")

# Split data into train and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X_combined, y, test_size=0.2, random_state=42, stratify=y
)

print(f"\n📈 Data split:")
print(f"  Training set: {X_train.shape[0]} samples")
print(f"  Test set: {X_test.shape[0]} samples")
```

```
🔧 Preparing features for classification...
✅ Feature matrix prepared!
📊 Feature matrix shape: (1490, 5007)
🎯 Number of classes: 5
📋 Classes: ['business' 'entertainment' 'politics' 'sport' 'tech']

📈 Data split:
  Training set: 1192 samples
  Test set: 298 samples
```

```
import numpy as np
# Train and evaluate multiple classifiers
print("🤖 Training multiple classifiers...")

# Define classifiers to compare
classifiers = {
    'Naive Bayes': MultinomialNB(),
    'Logistic Regression': LogisticRegression(random_state=42, max_iter=1000),
    'SVM': SVC(random_state=42, probability=True)  # Enable probability for bet
}

# 💡 TIP: For larger datasets, you might want to use SGDClassifier for efficie
# from sklearn.linear_model import SGDClassifier
# classifiers['SGD'] = SGDClassifier(random_state=42)

# Train and evaluate each classifier
results = {}
trained_models = {}

# Determine the index of the 'full_sentiment' feature in the combined array
# X_tfidf.shape[1] gives the number of TF-IDF features.
# The 'full_sentiment' feature is the first of the sentiment features immediate
```

```
full_sentiment_col_idx = tfidf_matrix.shape[1]

for name, classifier in classifiers.items():
    print(f"\n🔄 Training {name}...")

    # Prepare features based on classifier type
    current_X_train = X_train
    current_X_test = X_test

    if isinstance(classifier, MultinomialNB):
        # MultinomialNB requires non-negative features. Scale 'full_sentiment'
        current_X_train = X_train.copy()
        current_X_test = X_test.copy()
        current_X_train[:, full_sentiment_col_idx] = (current_X_train[:, full_s
        current_X_test[:, full_sentiment_col_idx] = (current_X_test[:, full_ser

    # 🚀 YOUR CODE HERE: Train and evaluate classifier
    # Train the model
    classifier.fit(current_X_train, y_train)

    # Make predictions
    y_pred = classifier.predict(current_X_test)
    y_pred_proba = classifier.predict_proba(current_X_test) if hasattr(classifi

    # Calculate metrics
    accuracy = accuracy_score(y_test, y_pred)

    # Cross-validation score
    cv_scores = cross_val_score(classifier, current_X_train, y_train, cv=3, scc

    # Store results
    results[name] = {
        'accuracy': accuracy,
        'cv_mean': cv_scores.mean(),
        'cv_std': cv_scores.std(),
        'predictions': y_pred,
        'probabilities': y_pred_proba
    }

    trained_models[name] = classifier

    print(f"  ✅ Accuracy: {accuracy:.4f}")
    print(f"  📊 CV Score: {cv_scores.mean():.4f} (+/- {cv_scores.std() * 2:.4

print("\n🏆 CLASSIFIER COMPARISON")
print("=" * 50)
comparison_df = pd.DataFrame({
    'Model': list(results.keys()),
    'Test Accuracy': [results[name]['accuracy'] for name in results.keys()],
    'CV Mean': [results[name]['cv_mean'] for name in results.keys()],
    'CV Std': [results[name]['cv_std'] for name in results.keys()]
```

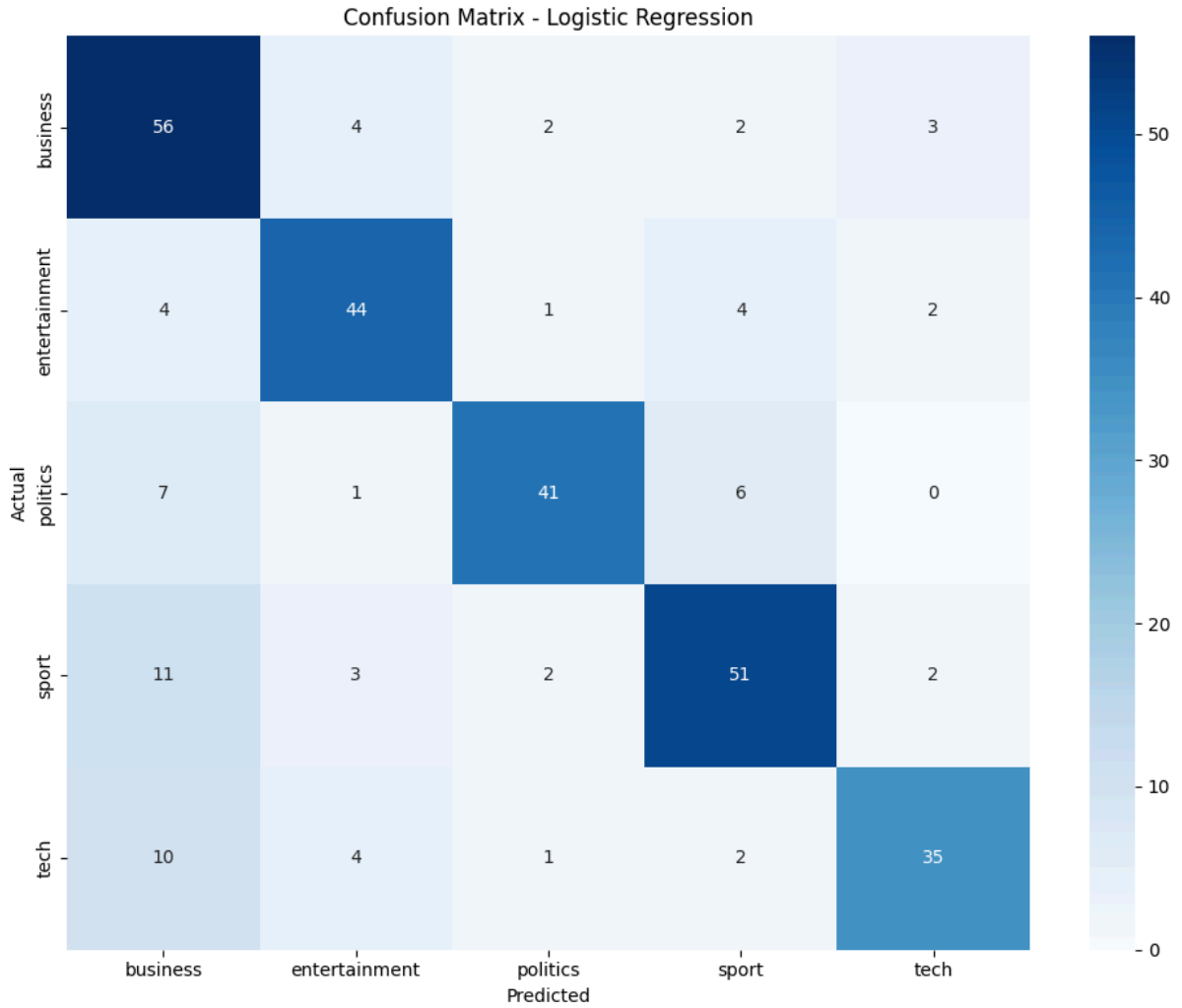
```
})

print(comparison_df.round(4))

# Find best model
best_model_name = comparison_df.loc[comparison_df['Test Accuracy'].idxmax(), 'M
print(f"\n🥇 Best performing model: {best_model_name}")
```

```
🤖 Training multiple classifiers...

🔄 Training Naive Bayes...
  ✅ Accuracy: 0.6141
  📊 CV Score: 0.5503 (+/- 0.0498)

🔄 Training Logistic Regression...
  ✅ Accuracy: 0.7617
  📊 CV Score: 0.7610 (+/- 0.1762)

🔄 Training SVM...
  ✅ Accuracy: 0.3557
  📊 CV Score: 0.3574 (+/- 0.0404)

🏆 CLASSIFIER COMPARISON
==================================================
                Model  Test Accuracy  CV Mean  CV Std
0         Naive Bayes         0.6141   0.5503  0.0249
1  Logistic Regression         0.7617   0.7610  0.0881
2                  SVM         0.3557   0.3574  0.0202

🥇 Best performing model: Logistic Regression
```

```
# Detailed evaluation of the best model
best_model = trained_models[best_model_name]
best_predictions = results[best_model_name]['predictions']

print(f"📊 DETAILED EVALUATION: {best_model_name}")
print("=" * 60)

# Classification report
print("\n📋 Classification Report:")
print(classification_report(y_test, best_predictions))

# Confusion matrix
cm = confusion_matrix(y_test, best_predictions)
plt.figure(figsize=(10, 8))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=np.unique(y), yticklabels=np.unique(y))
plt.title(f'Confusion Matrix - {best_model_name}')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.tight_layout()
plt.show()
```

```
# Feature importance (for Logistic Regression)
if best_model_name == 'Logistic Regression':
    print("\n🔍 Top Features by Category:")
    feature_names_extended = list(feature_names) + ['sentiment', 'pos_score', '
                                                    'char_length', 'word_count',

    classes = best_model.classes_
    coefficients = best_model.coef_

    for i, class_name in enumerate(classes):
        top_indices = np.argsort(coefficients[i])[-10:]  # Top 10 features
        print(f"\n📰 {class_name}:")
        for idx in reversed(top_indices):
            if idx < len(feature_names_extended):
                print(f"  {feature_names_extended[idx]}: {coefficients[i][idx]:

# 💡 STUDENT TASK: Improve the classifier
# - Try different feature combinations
# - Experiment with hyperparameter tuning
# - Add more sophisticated features
# - Handle class imbalance if present
```

```
📊 DETAILED EVALUATION: Logistic Regression
============================================================

📋 Classification Report:
               precision    recall  f1-score   support

     business       0.64      0.84      0.72        67
entertainment       0.79      0.80      0.79        55
     politics       0.87      0.75      0.80        55
        sport       0.78      0.74      0.76        69
         tech       0.83      0.67      0.74        52

     accuracy                           0.76       298
    macro avg       0.78      0.76      0.77       298
 weighted avg       0.78      0.76      0.76       298
```




```
🔍 Top Features by Category:

📰 business:
  neu_score: 1.6171
  bank: 1.3510
  economy: 1.2481
  firm: 1.2328
  growth: 1.2120
  market: 1.1905
```

```
company: 1.1002
price: 1.0875
rate: 1.0708
share: 1.0676
🎬 entertainment:
film: 3.2701
award: 1.6436
sentiment: 1.5821
best: 1.3570
band: 1.2379
star: 1.1697
actor: 1.1610
album: 1.1524
show: 1.1147
🏛️ politics:
labour: 2.3322
election: 2.1445
party: 2.0346
blair: 1.9865
tory: 1.5266
minister: 1.2403
```

🔍 Named Entity Recognition

🎯 Module 8: Extracting Facts from News

Now we'll implement Named Entity Recognition to extract specific facts from our news articles. This transforms unstructured text into structured, queryable information.

NER Applications:

- **Entity Tracking:** Monitor mentions of people, organizations, locations
- **Fact Extraction:** Build knowledge bases from news content
- **Relationship Mapping:** Understand connections between entities
- **Timeline Construction:** Track events and their participants

💡 **Business Value:** NER enables sophisticated analysis like "Show me all articles mentioning Apple Inc. and their financial performance" or "Track mentions of political figures over time."

```
def extract_entities(text):
    """
    Extract named entities using spaCy

    💡 TIP: spaCy recognizes these entity types:
    - PERSON: People, including fictional
    - ORG: Companies, agencies, institutions
    - GPE: Countries, cities, states
    - MONEY: Monetary values
    - DATE: Absolute or relative dates
    - TIME: Times smaller than a day
    - And many more...
    """
    if not text or pd.isna(text):
        return []

    # 🚀 YOUR CODE HERE: Implement entity extraction
    doc = nlp(str(text))

    entities = []
    for ent in doc.ents:
        entities.append({
            'text': ent.text,
            'label': ent.label_,
            'start': ent.start_char,
            'end': ent.end_char,
            'description': spacy.explain(ent.label_)
        })
```

```
    return entities

# Apply NER to all articles
print("🔍 Extracting named entities...")

all_entities = []
article_entities = []

for idx, row in df.iterrows():
    entities = extract_entities(row['full_text'])

    # Store entities for this article
    article_entities.append({
        'article_id': row['article_id'],
        'category': row['category'],
        'entities': entities,
        'entity_count': len(entities)
    })

    # Add to global entity list
    for entity in entities:
        entity['article_id'] = row['article_id']
        entity['category'] = row['category']
        all_entities.append(entity)

print(f"✅ Entity extraction complete!")
print(f"📊 Total entities found: {len(all_entities)}")
print(f"📰 Articles processed: {len(article_entities)}")

# Convert to DataFrame for analysis
entities_df = pd.DataFrame(all_entities)

if not entities_df.empty:
    print(f"\n🏷️ Entity types found: {entities_df['label'].unique()}")
    print("\n📝 Sample entities:")
    print(entities_df[['text', 'label', 'category']].head(10))
else:
    print("⚠️ No entities found. This might happen with very short sample text
```

```
🔍 Extracting named entities...
✅ Entity extraction complete!
📊 Total entities found: 43706
📰 Articles processed: 1490

🏷️ Entity types found: ['ORDINAL' 'PERSON' 'GPE' 'DATE' 'MONEY' 'ORG' 'NORP' 'LO
 'PERCENT' 'TIME' 'EVENT' 'QUANTITY' 'FAC' 'PRODUCT' 'LANGUAGE'
 'WORK_OF_ART' 'LAW']

📝 Sample entities:
             text    label  category
0           first  ORDINAL  business
1   cynthia cooper   PERSON  business
```

```
2              us      GPE  business
3            2002     DATE  business
4           5.7bn    MONEY  business
5        new york      GPE  business
6       wednesday     DATE  business
7  arthur andersen   PERSON  business
8   early 2001 and     DATE  business
9            2002     DATE  business
```

```
# Analyze entity patterns
if not entities_df.empty:
    print("📊 NAMED ENTITY ANALYSIS")
    print("=" * 50)

    # Entity type distribution
    entity_counts = entities_df['label'].value_counts()
    print("\n🏷️ Entity type distribution:")
    print(entity_counts)

    # Entity types by category
    entity_by_category = entities_df.groupby(['category', 'label']).size().unst
    print("\n📰 Entity types by news category:")
    print(entity_by_category)

    # Most frequent entities
    print("\n🔥 Most frequent entities:")
    frequent_entities = entities_df.groupby(['text', 'label']).size().sort_valu
    for (entity, label), count in frequent_entities.items():
        print(f"  {entity} ({label}): {count} mentions")

    # Visualizations
    fig, axes = plt.subplots(2, 2, figsize=(15, 12))

    # 1. Entity type distribution
    entity_counts.plot(kind='bar', ax=axes[0,0])
    axes[0,0].set_title('Entity Type Distribution')
    axes[0,0].tick_params(axis='x', rotation=45)

    # 2. Entities per category
    entities_per_category = entities_df.groupby('category').size()
    entities_per_category.plot(kind='bar', ax=axes[0,1])
    axes[0,1].set_title('Total Entities per Category')
    axes[0,1].tick_params(axis='x', rotation=45)

    # 3. Entity type heatmap by category
    if entity_by_category.shape[0] > 1 and entity_by_category.shape[1] > 1:
        sns.heatmap(entity_by_category, annot=True, fmt='d', ax=axes[1,0], cmap
        axes[1,0].set_title('Entity Types by Category Heatmap')
    else:
        axes[1,0].text(0.5, 0.5, 'Insufficient data\nfor heatmap',
                       ha='center', va='center', transform=axes[1,0].transAxes)
```

```
            axes[1,0].set_title('Entity Types by Category')

        # 4. Top entities
        top_entities = entities_df['text'].value_counts().head(10)
        top_entities.plot(kind='barh', ax=axes[1,1])
        axes[1,1].set_title('Most Mentioned Entities')

        plt.tight_layout()
        plt.show()

        # 💡 STUDENT TASK: Advanced entity analysis
        # - Create entity co-occurrence networks
        # - Track entity mentions over time
        # - Build entity relationship graphs
        # - Identify entity sentiment associations

    else:
        print("⚠️ Skipping entity analysis due to insufficient data.")
        print("💡 TIP: Try with a larger, more diverse dataset for better NER resu
```

```
📊 NAMED ENTITY ANALYSIS
==================================================

🏷️ Entity type distribution:
label
PERSON         8923
DATE           8851
CARDINAL       6321
GPE            5805
ORG            4495
NORP           2923
MONEY          1786
ORDINAL        1683
PERCENT        1276
TIME            593
LOC             524
QUANTITY        199
FAC              94
PRODUCT          88
LANGUAGE         62
EVENT            43
LAW              33
WORK_OF_ART       7
Name: count, dtype: int64

📰 Entity types by news category:
label          CARDINAL  DATE  EVENT  FAC   GPE  LANGUAGE  LAW  LOC  MONEY  \
category
business           1066  2429      3   10  1655         1   15  175    907
entertainment      1130  1745     12   15   808         9    4   65    386
politics            845  1362     22   27   963        10    4  122    235
sport              1839  1991      5   40  1592        12    5   38     44
tech               1441  1324      1    2   787        30    5  124    214

label          NORP  ORDINAL   ORG  PERCENT  PERSON  PRODUCT  QUANTITY  TIME
category
business        728      178  1424      813    1065       22        42    53
entertainment   451      349   656       43    2030        8        20   128
politics        746      223   586      125    2141        4        12    82
sport           695      669   680       18    2920       25        65   240
tech            303      264   449      272     767       29        60    90

label          WORK_OF_ART
category
business                 0
entertainment            0
politics                 1
sport                    1
tech                     5

🔥 Most frequent entities:
  first (CARDINAL): 770 mentions
  one (CARDINAL): 680 mentions
  two (CARDINAL): 612 mentions
  uk (GPE): 565 mentions
  us (GPE): 551 mentions
```

📈 Comprehensive Analysis and Insights

🎯 Bringing It All Together

Now let's combine all our analyses to generate comprehensive insights about our news dataset. This is where the real business value emerges from our NLP pipeline.

Key Analysis Areas:

1. **Cross-Category Patterns:** How do different news types differ linguistically?
2. **Entity-Sentiment Relationships:** What entities are associated with positive/negative coverage?

3. **Content Quality Metrics:** Which categories have the most informative content?
4. **Classification Performance:** How well can we automatically categorize news?

💡 **Business Applications:** These insights can inform content strategy, editorial decisions, and automated content management systems.

```
british (NORP): 349 mentions
second (ORDINAL): 329 mentions
three (CARDINAL): 328 mentions
britain (GPE): 295 mentions
london (GPE): 282 mentions
2004 (DATE): 267 mentions
france (GPE): 255 mentions
2005 (DATE): 247 mentions
```

```
# Create comprehensive analysis dashboard
def create_comprehensive_analysis():
    """
    Generate comprehensive insights combining all analyses

    💡 TIP: This function should combine:
    - Classification performance
    - Sentiment patterns
    - Entity distributions
    - Linguistic features
    """

    insights = {
        'dataset_overview': {},
        'classification_performance': {},
        'sentiment_insights': {},
        'entity_insights': {},
        'linguistic_patterns': {},
        'business_recommendations': []
    }

    # 🚀 YOUR CODE HERE: Generate comprehensive insights

    # Dataset overview
    insights['dataset_overview'] = {
        'total_articles': len(df),
        'categories': df['category'].unique().tolist(),
        'category_distribution': df['category'].value_counts().to_dict(),
        'avg_article_length': df['full_text'].str.len().mean(),
        'avg_words_per_article': df['full_text'].str.split().str.len().mean()
    }

    # Classification performance
    insights['classification_performance'] = {
        'best_model': best_model_name,
        'best_accuracy': results[best_model_name]['accuracy'],
        'model_comparison': {name: results[name]['accuracy'] for name in result
    }

    # Sentiment insights
    sentiment_by_cat = sentiment_df.groupby('category')['full_sentiment'].mean(
    insights['sentiment_insights'] = {
        'most_positive_category': max(sentiment_by_cat, key=sentiment_by_cat.ge
        'most_negative_category': min(sentiment_by_cat, key=sentiment_by_cat.ge
```

```
            'sentiment_by_category': sentiment_by_cat,
            'overall_sentiment': sentiment_df['full_sentiment'].mean()
        }

        # Entity insights
        if not entities_df.empty:
            entity_by_cat = entities_df.groupby('category').size().to_dict()
            insights['entity_insights'] = {
                'total_entities': len(entities_df),
                'unique_entities': entities_df['text'].nunique(),
                'entity_types': entities_df['label'].unique().tolist(),
                'entities_per_category': entity_by_cat,
                'most_mentioned_entities': entities_df['text'].value_counts().head(
            }

        # Generate business recommendations
        recommendations = []

        # Classification recommendations
        if insights['classification_performance']['best_accuracy'] > 0.8:
            recommendations.append("✅ High classification accuracy achieved - rea
        else:
            recommendations.append("⚠️ Classification accuracy needs improvement -

        # Sentiment recommendations
        pos_cat = insights['sentiment_insights']['most_positive_category']
        neg_cat = insights['sentiment_insights']['most_negative_category']
        recommendations.append(f"📊 {pos_cat} articles are most positive - good fo
        recommendations.append(f"📊 {neg_cat} articles are most negative - may nee

        # Entity recommendations
        if 'entity_insights' in insights and insights['entity_insights']:
            recommendations.append("🔍 Rich entity extraction enables advanced sea

        insights['business_recommendations'] = recommendations

        return insights

    # Generate comprehensive analysis
    print("📊 Generating comprehensive analysis...")
    analysis_results = create_comprehensive_analysis()

    print("✅ Analysis complete!")
    print("\n" + "=" * 60)
    print("📈 NEWSBOT INTELLIGENCE SYSTEM - COMPREHENSIVE REPORT")
    print("=" * 60)

    # Display key insights
    print(f"\n📊 DATASET OVERVIEW:")
    overview = analysis_results['dataset_overview']
    print(f"  Total Articles: {overview['total_articles']}")
```

```
print(f"  Categories: {', '.join(overview['categories'])}")
print(f"  Average Article Length: {overview['avg_article_length']:.0f} characte
print(f"  Average Words per Article: {overview['avg_words_per_article']:.0f} wo

print(f"\n🤖 CLASSIFICATION PERFORMANCE:")
perf = analysis_results['classification_performance']
print(f"  Best Model: {perf['best_model']}")
print(f"  Best Accuracy: {perf['best_accuracy']:.4f}")

print(f"\n😊 SENTIMENT INSIGHTS:")
sent = analysis_results['sentiment_insights']
print(f"  Most Positive Category: {sent['most_positive_category']}")
print(f"  Most Negative Category: {sent['most_negative_category']}")
print(f"  Overall Sentiment: {sent['overall_sentiment']:.4f}")

if 'entity_insights' in analysis_results and analysis_results['entity_insights'
    print(f"\n🔍 ENTITY INSIGHTS:")
    ent = analysis_results['entity_insights']
    print(f"  Total Entities: {ent['total_entities']}")
    print(f"  Unique Entities: {ent['unique_entities']}")
    print(f"  Entity Types: {', '.join(ent['entity_types'])}")

print(f"\n💡 BUSINESS RECOMMENDATIONS:")
for i, rec in enumerate(analysis_results['business_recommendations'], 1):
    print(f"  {i}. {rec}")
```

```
📊 Generating comprehensive analysis...
✅ Analysis complete!

============================================================
📈 NEWSBOT INTELLIGENCE SYSTEM - COMPREHENSIVE REPORT
============================================================

📊 DATASET OVERVIEW:
  Total Articles: 1490
  Categories: business, tech, politics, sport, entertainment
  Average Article Length: 2297 characters
  Average Words per Article: 386 words

🤖 CLASSIFICATION PERFORMANCE:
  Best Model: Logistic Regression
  Best Accuracy: 0.7617

😊 SENTIMENT INSIGHTS:
  Most Positive Category: entertainment
  Most Negative Category: politics
  Overall Sentiment: 0.3950

🔍 ENTITY INSIGHTS:
  Total Entities: 43706
  Unique Entities: 12410
  Entity Types: ORDINAL, PERSON, GPE, DATE, MONEY, ORG, NORP, LOC, CARDINAL, PER

💡 BUSINESS RECOMMENDATIONS:
```

```
1. ⚠️ Classification accuracy needs improvement - consider more training data
2. 📊 entertainment articles are most positive - good for uplifting content r
3. 📊 politics articles are most negative - may need balanced coverage monito
4. 🔍 Rich entity extraction enables advanced search and relationship analysi
```

🚀 Final System Integration

🎯 Building the Complete NewsBot Pipeline

Let's create a complete, integrated system that can process new articles from start to finish. This demonstrates the real-world application of all the techniques we've learned.

Complete Pipeline:

1. **Text Preprocessing:** Clean and normalize input
2. **Feature Extraction:** Generate TF-IDF and other features
3. **Classification:** Predict article category
4. **Entity Extraction:** Identify key facts
5. **Sentiment Analysis:** Determine emotional tone
6. **Insight Generation:** Provide actionable intelligence

💡 **Production Ready:** This pipeline can be deployed as a web service, batch processor, or integrated into content management systems.

```
class NewsBotIntelligenceSystem:
    """
    Complete NewsBot Intelligence System

    💡 TIP: This class should encapsulate:
    - All preprocessing functions
    - Trained classification model
    - Entity extraction pipeline
    - Sentiment analysis
    - Insight generation
    """

    def __init__(self, classifier, vectorizer, sentiment_analyzer):
        self.classifier = classifier
        self.vectorizer = vectorizer
        self.sentiment_analyzer = sentiment_analyzer
        self.nlp = nlp  # spaCy model

    def preprocess_article(self, title, content):
        """Preprocess a new article"""
        full_text = f"{title} {content}"
        processed_text = preprocess_text(full_text)
```

```
        return full_text, processed_text

    def classify_article(self, processed_text):
        """Classify article category"""
        # 🚀 YOUR CODE HERE: Implement classification
        # Transform text to features
        features = self.vectorizer.transform([processed_text])

        # Add dummy features for sentiment and length (in production, calculate
        dummy_features = np.zeros((1, 7))  # 4 sentiment + 3 length features
        features_combined = np.hstack([features.toarray(), dummy_features])

        # Predict category and probability
        prediction = self.classifier.predict(features_combined)[0]
        probabilities = self.classifier.predict_proba(features_combined)[0]

        # Get class probabilities
        class_probs = dict(zip(self.classifier.classes_, probabilities))

        return prediction, class_probs

    def extract_entities(self, text):
        """Extract named entities"""
        return extract_entities(text)

    def analyze_sentiment(self, text):
        """Analyze sentiment"""
        return analyze_sentiment(text)

    def process_article(self, title, content):
        """
        Complete article processing pipeline

        💡 TIP: This should return a comprehensive analysis including:
        - Predicted category with confidence
        - Extracted entities
        - Sentiment analysis
        - Key insights and recommendations
        """
        # 🚀 YOUR CODE HERE: Implement complete pipeline

        # Step 1: Preprocess
        full_text, processed_text = self.preprocess_article(title, content)

        # Step 2: Classify
        category, category_probs = self.classify_article(processed_text)

        # Step 3: Extract entities
        entities = self.extract_entities(full_text)

        # Step 4: Analyze sentiment
```

```
            sentiment = self.analyze_sentiment(full_text)

            # Step 5: Generate insights
            insights = self.generate_insights(category, entities, sentiment, categc

            return {
                'title': title,
                'content': content[:200] + '...' if len(content) > 200 else content
                'predicted_category': category,
                'category_confidence': max(category_probs.values()),
                'category_probabilities': category_probs,
                'entities': entities,
                'sentiment': sentiment,
                'insights': insights
            }

        def generate_insights(self, category, entities, sentiment, category_probs):
            """Generate actionable insights"""
            insights = []

            # Classification insights
            confidence = max(category_probs.values())
            if confidence > 0.8:
                insights.append(f"✅ High confidence {category} classification ({c
            else:
                insights.append(f"⚠️ Uncertain classification - consider manual re

            # Sentiment insights
            if sentiment['compound'] > 0.1:
                insights.append(f"😊 Positive sentiment detected ({sentiment['comp
            elif sentiment['compound'] < -0.1:
                insights.append(f"😞 Negative sentiment detected ({sentiment['comp
            else:
                insights.append(f"😐 Neutral sentiment ({sentiment['compound']:.3f

            # Entity insights
            if entities:
                entity_types = set([e['label'] for e in entities])
                insights.append(f"🔍 Found {len(entities)} entities of {len(entity_

                # Highlight important entities
                important_entities = [e for e in entities if e['label'] in ['PERSON
                if important_entities:
                    key_entities = [e['text'] for e in important_entities[:3]]
                    insights.append(f"🎯 Key entities: {', '.join(key_entities)}")
            else:
                insights.append("ℹ️ No named entities detected")

            return insights

    # Initialize the complete system
```